

Computer and Network Security, Summer 2023

Project : Ransomware

Due: Saturday September 2nd by 23:59

1. Introduction

In this project you are required to develop a ransomware using python. This can be done using encryption and client-server architecture in order to be able to exchange data between the attacker and the victim.

2. Overview Your main goal is to write a malware that does the following:

- Generate a random 128 bit key (16 characters) using ascii characters.
- Find all .txt files in the Documents directory on the victim's computer and encrypt the bytes of these files.
- Send that key back to the server.
- Infect by sending your compiled malware to a list of emails.

3. Requirements

- Payload
 - a) Encrypt ALL .txt files in the documents directory and sub directories.
 - b) Use AES (choose any mode) for the encryption process with the random key mentioned at the beginning.
 - c) Generate a Public/Private key-pair using RSA (server side)
 - d) Save the key used in (b) to desktop in a .key (normal text file but change the extension to .key not to mistakenly re-encrypt it) format named "Key.key".
 - e) Encrypt the key used in (b) using the public key then save the encrypted key to the desktop in a .key file format named "encryptedKey.key".
 - f) Save the generated Public/Private key pair (one key per line, Public then Private) in another .key file named "keyPair.key".
 - g) Send the encrypted key from (e) to the server.
 - h) A decryption function must be present and will decrypt all files when the original key is entered.
- Infection
 - a) You should add an infection mechanism that sends the .exe file to the emails in the following CSV file.
 - b) You need to include your ransomware (hidden) in your email.
 - c) Your code should access the following csv , extract the emails from it then send the .exe file to them
<https://docs.google.com/spreadsheets/d/1shIs0NOFWPRPipMa7kzKKqQQ3D2Gki0VtaIMg0w3a8c/edit#gid=0>(do not hardcode the emails)

- Interface
 - a) A prompt should popup (CLI or GUI) indicating that the encryption is in progress when the .exe is executed.
 - b) After encrypting all txt files, the prompt will wait for a key input in order to decrypt the files back. Enter Key to decrypt"

4. Setup

- Your code should include a server side as well as a client side.
- After setting your environment up, you are required to download the "Starter-Code.zip" folder from the CMS, extract it and write your implementation in "Client.py" file.
- After writing your malware, use auto-py-to-exe library to convert Client.py file into a .exe file then copy this file into the virtual machine in order to be able to run the ransomware.

5. Submission

- a) Your finished ransomware must be submitted as a working .exe file
- b) A Report with team name and members w/ ID and tutorial number.
- c) The report must include a block diagram for the encryption/decryption along with their functions.
- d) Zip the report, your .exe file and source code all in one file to submit it.

6. IMPORTANT NOTES

- a) If any of windows security components blocks your malware executable: disable them. (Google is your friend)
- b) You should implement your work in python.
- c) Google is your friend.
- d) You must stick to the naming conventions of the files since they will be automatically graded
- e) IP addresses should be dynamic.
- f) Make sure your email is configured properly.

7. Important Dates

Team Submission Make sure you submit your team details by Saturday 19th of August at 23:59 using the following link <https://forms.gle/kHnNKW4acPVHkfix7>

Only one team member should submit this for the whole team. After this deadline, we will be posting on the CMS a team ID for each submitted team. You will be using this team ID for submission.

Source code and Project Report On-line submission by September 2nd at 23:59. The submission details will be announced after the team submission deadline.